1 Question (10 points)

Modify Dijkstra's shortest path algorithm with nonnegative numbers assigned to its vertices so that the length of a path defined as the sum of the vertex numbers on the path.

2 Question (15 points)

Given the following function

```
int algorithm(int N){
   int sum = 0;
   if (N == 0)
       return 0;
   for (i = 0; i < N; i++)
      for (int j = 0; j < N; j++)
         sum++;
   return algorithm(N - 1) + sum;
}
```

What is the time complexity of the function algorithm? Show all your work. Derive the recursive equation and solve it.

3 Question (15 points)

A word can be changed to another word by a 1-character substitution. Assume that a dictionary of 5-letter words exists. Give an algorithm to determine if a word A can be transformed to a word B by a series of 1-character substitutions, and if so, outputs the corresponding sequence of words. Note that all intermediate words must be in the dictionary. As an example, bleed converts to blood by the sequence bleed, blend, blond, blood. Convert this problem into a graph problem and solve with an appropriate algorithm.

4 Question (10 points)

The swap sorting of permutation π is a transformation of π into the identity permutation by exchanges of adjacent elements. For example, 3142 $\rightarrow$ 1342 $\rightarrow$ 1324 $\rightarrow$ 1234 is a three-step swap sorting of permutation 1234. Write a greedy algorithm for swap sorting that uses the minimum number of swaps to sort a permutation.